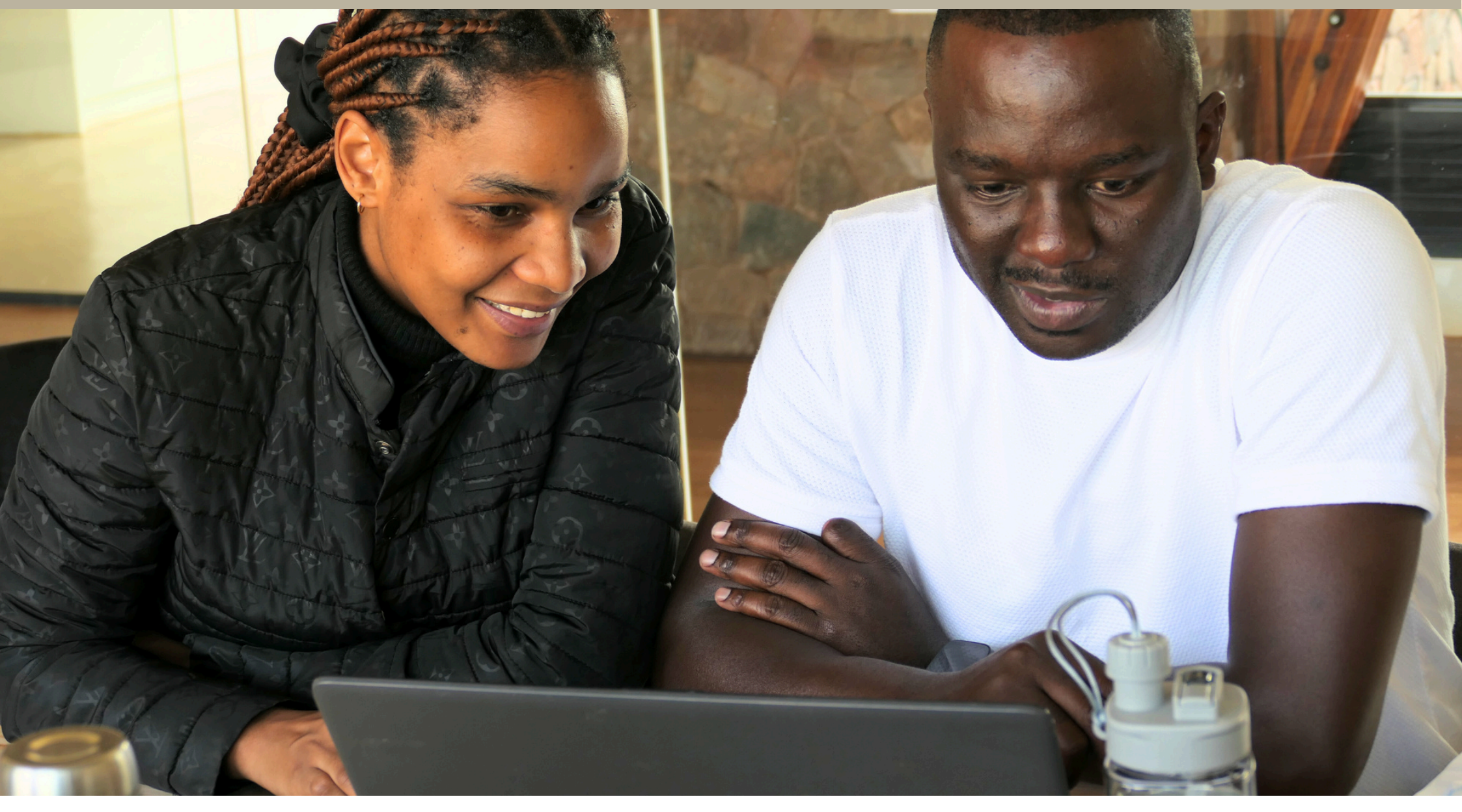
A Call for Applications
Advanced GIS Training-AGT2025



Training overview

The Grumeti Fund Trust through Research and Innovation for the Serengeti Ecosystem (RISE) will be hosting an advanced GIS training for early to mid-career conservation practitioners from July 14 to 19, 2025 (Six days).

The Advanced GIS Training (AGT2025) is being organized by collaborators from RISE, Pennsylvania State University, Montana State University Billings, the University of Connecticut, and the University of Dar es Salaam. Fifteen (15) candidates will be selected for in-person participation at the RISE facility at Grumeti, Tanzania, and twenty five (25) for online participation. The training will delve into advanced topics in GIS, mainly covering applications of important geospatial methods in the analysis of climate and environmental data. This training is designed for professionals, researchers, and enthusiasts who have some basic background in GIS and would like to deepen their knowledge of geospatial data analysis and applications in analyzing environmental issues.

The training will center on the use of advanced geospatial analysis methods to analyze freely available climate and environmental data. Modules to be covered include GIS fundamentals for ecologists, landscape pattern analysis in R, spatial analysis of ecological data in R, spatial data sources and access, landscape analysis in R and Google Earth Engine (GEE), and habitat connectivity. A free ArcGIS Pro license valid for one year will be provided to each participant who does not have access to the software. However, participants should bring their own laptops during the training.



Training objectives

By the end of the training, participants should be able to:

1. Identify, download, and process open-source climate and environmental data
2. Perform analysis of climate, environmental, and ecological data in commercial GIS software (ArcGIS Pro) and open-source software (R and Google Earth Engine)
3. Apply GIS and spatial statistics to analyze landscape structure and pattern
4. Apply GIS and multi-criteria analysis techniques to map habitat connectivity for selected species.
5. Apply time series analysis to assess patterns and changes in important landscape characteristics and functions
6. Present, in the form of maps and figures, results from their geospatial analysis

Delivery Format

A typical training day will consist of three sections: self-learning, group work, and live sessions. Two instructors will be on-site, and three will deliver their materials remotely. Participants are expected to take charge of their own learning experience.

Eligibility

1. Persons with a training background in the fields of ecology, agriculture, natural resource management, geography, geoinformatics, zoology, botany, forestry, and related fields
2. Should be able to perform basic spatial analysis tasks. Prior experience with GIS tools (especially ArcGIS Pro or ArcMap) is a plus. Otherwise, learners will be required to familiarize themselves with the ArcGIS software during the self-learning sessions
3. Be familiar with basic GIS concepts, e.g., coordinate systems, GIS data models, geoprocessing, and map making
4. Should be enthusiastic about GIS and seeking to advance their knowledge, and planning to apply the learned skills in their work and/or research
5. Participants should be ready and willing to commit the entire six days to this training. They should be willing to participate actively in live sessions, self-learning activities, and group projects assigned during the training period
6. Present, in the form of maps and figures, results from their geospatial analysis

How to Apply

To apply for this training, please fill Google Form and if you have any difficulty during the application, please send an email to RISE@grumetifund.org, the email subject line should be Advanced GIS Training.

Deadline: The deadline for application will be May 25th, 2025, before 2359 hrs EAT.

NOTE: This training will be offered with the support from the Grumeti Fund. There will be no direct cost to the selected on-site participants. We will reimburse all ground travel expenses (not flights) and provide meals and accommodation for those participating on-site. There will be NO stipend/per diem provided during this training. Selected candidates will be informed by Early June 2025

Meet our instructors



Dr. Dan Wanyama
UCONN, USA



Dr. Olipa Simon
UDSM, TZ



Dr. Donald Akanga
MSUB, USA



Dr. Susan Kotikot
PSU, USA



Dr. Grace Malley
UDSM, TZ

Dr. Wanyama: Assistant Professor, Department of Geography, Sustainability, Urban, and Community Studies University of Connecticut

Dr. Olipa: Leading Scientist, Institute of Resource Assessment, University of Dar es salaam

Dr. Akanga: Assistant Professor, Environmental Studies Program Department of Social Sciences & Cultural Studies Montana State University Billings

Dr. Kotikot: Postdoctoral Fellow, Geography Department, Pennsylvania State University

Dr. Malley: Lecturer, Institute of Resource Assessment, University of Dar es salaam